Supplementary Material - Lab Documentation

Directions for students

Prepare $K_3[Fe(C_2O_4)_3]$ following the classic method of Yost and Simons (*7*).

Prepare an approximately 0.1 M solution (5 g in 100 mL of distilled water) of $K_3[Fe(C_2O_4)_3]\cdot 3H_2O$. (*The crystals can take up to 30 minutes to dissolve, even with stirring. If the crystals of* $K_3[Fe(C_2O_4)_3]\cdot 3H_2O$ *are finely ground in a mortar, this will result in a much shorter dissolution time.*) This will be the starting solution (**1**). We strongly recommend that this solution be stored in the dark if it is not used immediately.

Place 1 mL of the green solution **1** in a test tube and add 3 drops[1] of 6 M HCl 6; the solution turns yellow-orange (**2**). To **2** add 3 drops of 0.5 M KSCN; the solution turns red[2] (**3**). To **3** add 10 drops of 3 M KF; the solution becomes almost colorless (**4**). To **4** add 15 drops of 2 M $K_2C_2O_4$, the green solution **1** is obtained again:

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[1] When performing this experiment in our lab, we have found that drop counting is the best way to keep track of the amount of reagent added. In Supplementary Scheme 1 we give the amounts of reagent added as equivalents, which gives a much better idea of how easily an equilibrium is established or displaced.

[2] Examples are given in references *8*, *9* and *10* of quantitative determination (mainly through UV-VIS spectroscopy) of colored complexes such as $[Fe(SCN)_3]$, $Fe_3[Fe(CN)_6]_2$ and others. The instructor may find them useful to put the experiment on a more quantitative footing.

Scheme 1

$$[Fe(C_2O_4)_3]^{2-}\ (\mathbf{1}) \xrightarrow{H^+} [Fe(H_2O)_6]^{3+}\ (\mathbf{2}) + H_2C_2O_4$$

$$[Fe(H_2O)_6]^{3+}\ (\mathbf{2}) \xrightarrow{SCN^-} [Fe(SCN)_3]\ (\mathbf{3})$$

$$[Fe(SCN)_3]\ (\mathbf{3}) \xrightarrow{F^-} [FeF_4]^-\ (\mathbf{4})$$

$$[FeF_4]^-\ (\mathbf{4}) \xrightarrow{C_2O_4^{2-}} [Fe(C_2O_4)_3]^{3-}\ (\mathbf{1})$$

Alternative routes between solutions **1** and **4** are:

a) Add 10 drops of 3 M KF to 1 mL of solution **1**. This generates the almost colorless solution **4**, which can be reversed back to solution **1** by the addition of 15 drops of 2 M $K_2C_2O_4$.

b) Add 5 drops of 0.5 M KSCN to 1 mL of solution **1**, the green solution turns yellow-green (**5**). To **5** add 10 drops of 3 M KF to obtain the colorless solution **4**, which, as has been stated previously, can be converted back to solution **1** by the addition of 15 drops of 2 M $K_2C_2O_4$:

Scheme 2

$$[Fe(C_2O_4)_3]^{2-}\ (\mathbf{1}) \underset{C_2O_4^{2-}}{\overset{F^-}{\rightleftarrows}} [FeF_4]^-\ (\mathbf{4})$$

$$[Fe(C_2O_4)_3]^{2-}\ (\mathbf{1}) \xrightarrow{SCN^-} [Fe(SCN)_3]\ (\mathbf{5})\ \text{(small amount)}$$

$$[Fe(SCN)_3]\ (\mathbf{5})\ \text{(small amount)} \xrightarrow{F^-} [FeF_4]^-\ (\mathbf{4})$$

Add 1 mL of 0.1 M NaOH to 1 mL of solution **1**. A brown precipitate (**A**) and a colorless solution (**6**) are obtained that can be separated by centrifugation or filtration. Shaking the brown solid **A** with 1 mL of 1 M $H_2C_2O_4$ gives the green solution **1** again. Treatment of the colorless solution **6** with 1 mL of 0.2 M $BaCl_2$ gives a white precipitate (**B**). After centrifugation or filtration, the white solid **B** is treated with 10 drops of 1 M $Fe_2(SO_4)_3$. Centrifugation gives the green solution **1** again and a white precipitate (**C**):

Scheme 3

$$[Fe(C_2O_4)_3]^{2-}\ (\mathbf{1}) \xrightarrow{OH^-} Fe(OH)_3\ (\mathbf{A}) + C_2O_4^{2-}\ (\mathbf{6})$$

$$Fe(OH)_3\ (\mathbf{A}) \xrightarrow{H_2C_2O_4} [Fe(C_2O_4)_3]^{3-}\ (\mathbf{1})$$

$$C_2O_4^{2-}\ (\mathbf{6}) \xrightarrow{Ba^{2+}} BaC_2O_4\ (\mathbf{B})$$

$$BaC_2O_4\ (\mathbf{B}) \xrightarrow{Fe_2(SO_4)_3} [Fe(C_2O_4)_3]^{3-}\ (\mathbf{1}) + BaSO_4\ (\mathbf{C})$$

Add 5 mL of 0.2 M $BaCl_2$ to 1 mL of solution **1**. A pale-green precipitate (**D**) appears that can be separated by centrifugation or filtration. Treatment of the pale-green solid **D** with 3 mL of distilled water and heating until complete dissolution of the solid gives a solution (**7**). Addition of 3 drops of 0.5 M K_2SO_4 to **7** gives the white precipitate **C** and the green solution **1**:

Scheme 4

$$[Fe(C_2O_4)_3]^{2-}\ (\mathbf{1}) \xrightarrow{Ba^{2+}} Ba_3[Fe(C_2O_4)_3]_2\ (\mathbf{D})$$

$$Ba_3[Fe(C_2O_4)_3]_2\ (\mathbf{D}) \xrightarrow{H_2O,\ \Delta} \{Ba^{2+} + [Fe(C_2O_4)_3]^{3-}\}\ (\mathbf{7})$$

$$\{Ba^{2+} + [Fe(C_2O_4)_3]^{3-}\}\ (\mathbf{7}) \xrightarrow{SO_4^{2-}} [Fe(C_2O_4)_3]^{3-}\ (\mathbf{1}) + BaSO_4\ (\mathbf{C})$$

Take 1 mL of *freshly* prepared solution **1** and test it for Fe^{2+} with 1 drop of 0.1 M $K_3[Fe(CN)_6]$. Place 20 mL of solution **1** in a 50-mL beaker and expose it to the light from a flashlight (or a 60 W bulb at 15 cm from the surface of the solution) and repeat the Fe^{2+} test every 90 seconds for 5 minutes with aliquots of 1 mL (keep the tests for the first 5 minutes in the test tube rack in order to observe the progress of the reaction). After 10 minutes of exposure to light repeat the Fe^{2+} test and compare it with the previous ones. Put the remaining solution under a mild vacuum (a suction flask with a stopper and a filtration pump will be sufficient) and observe the evolution of CO_2 bubbles from it:

Scheme 5

$$[Fe(C_2O_4)_3]^{2-}\ (\mathbf{1}) \xrightarrow{[Fe(CN)_6]^{3-}} \text{no reaction}$$

$$[Fe(C_2O_4)_3]^{2-}\ (\mathbf{1}) \xrightarrow{h\nu} [Fe(H_2O)_6]^{2+} + CO_2\uparrow \xrightarrow{[Fe(CN)_6]^{3-}} Fe_3[Fe(CN)_6]_2$$

*During all these experiments it is convenient to keep a test tube with 1 mL of solution **1**, to be used as a color reference. If necessary, different amounts of water can be added to this solution to dilute it. Keeping a color reference handy helps the students appreciate the color changes that take place during the reactions.*

Supplementary hazards

In the event students lack adequate knowledge of laboratory safety procedures and good laboratory practices (our first-year Inorganic Chemistry Laboratory students have all taken a previous mandatory course on General Chemistry Laboratory, where they learn all the necessary operational skills and safety procedures) or the number of students per instructor is too high (we average 10 students per instructor), the most potentially dangerous reagents can be avoided or substituted:

a) $BaCl_2$ can be substituted by $CaCl_2$. In this case, precipitates **B**, **C** and **D** will be CaC_2O_4, $CaSO_4$ and $Ca_3[Fe(C_2O_4)_3]_2$, respectively. The concentration of reagents must be increased to take into account the higher solubility of Ca(II) salts in comparison with Ba(II) ones. Because of this increase in solubility for Ca(II) salts, optimum results are obtained when using Ba(II) salts, which should be preferred to Ca(II) ones whenever possible.[3]

b) KF can be avoided but not substituted. As there is not a safer substitute reagent for fluoride with similar properties in aqueous solution, the only way is to suppress the reactions using KF. In this case Schemes 1 and 2 will be simplified accordingly.

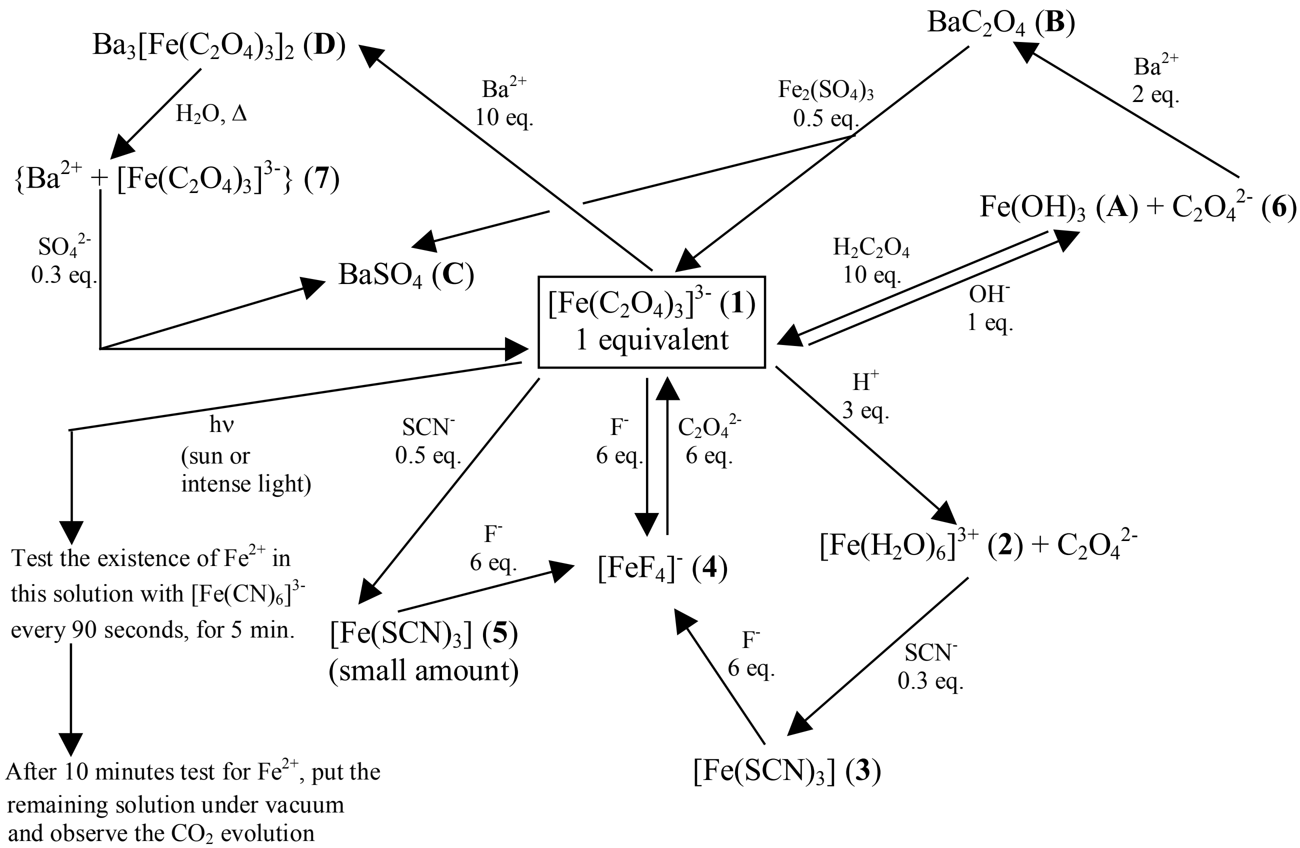
We leave to the instructor the final decision and choice of when to use these modifications, either individually or together.

[3] The use of Ba(II) salts provides us with a very good example of the importance of equilibrium and solubility. In each course we always emphasize to our students that Ba(II) salts as a general rule are toxic, but that highly insoluble $BaSO_4$ is harmless, which has allowed its use as a gastric contrast agent in X-ray examinations.

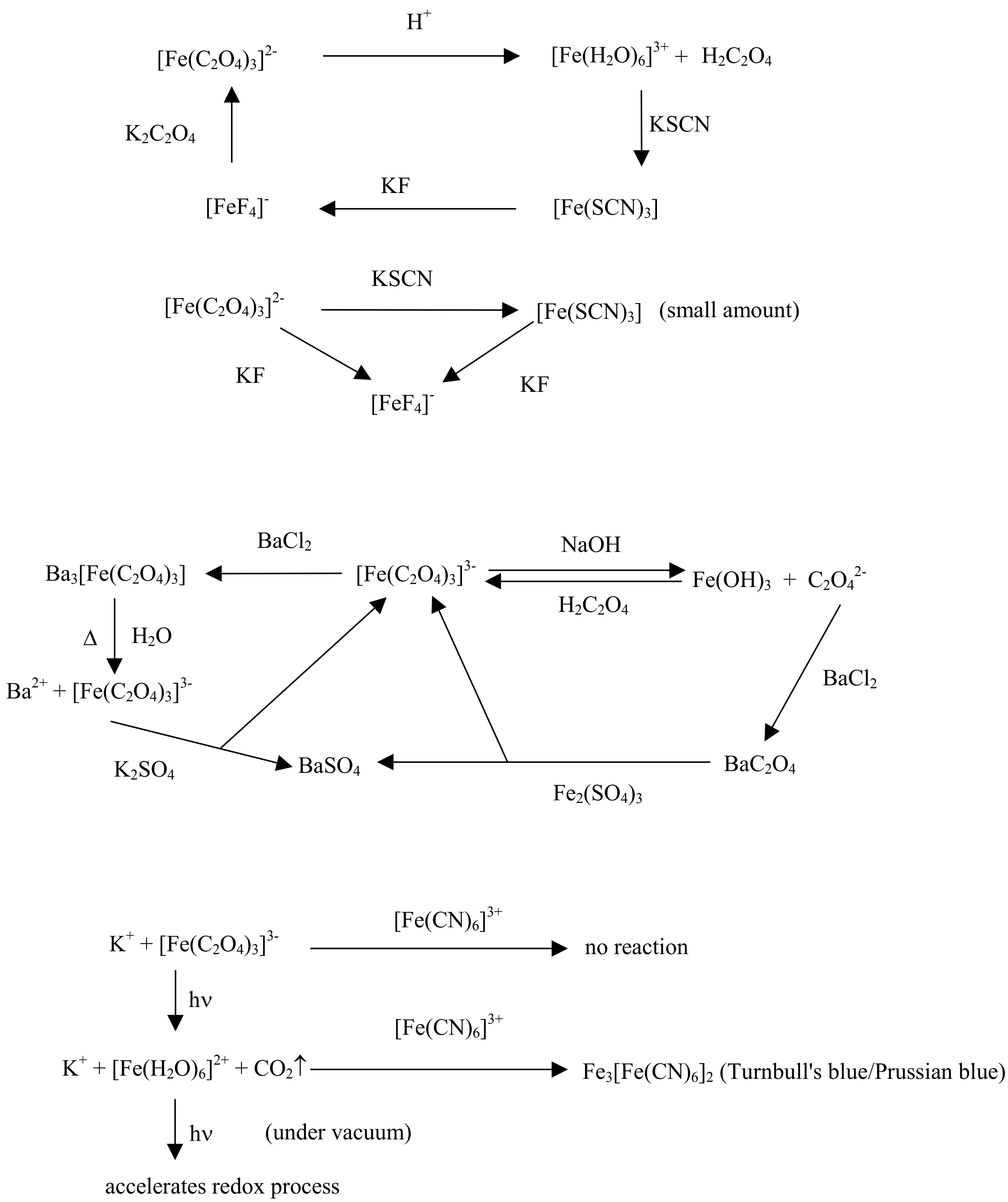
Supplementary assignment for students

After the experimental work is finished, instead of the set of questions, the students can be given a scheme to complete. This scheme (see Supplementary Scheme 3) summarizes all the experimental work (description of all the operations performed, reagents added and products formed). The students must identify solutions **2** through **7** and precipitates **A** through **D**.

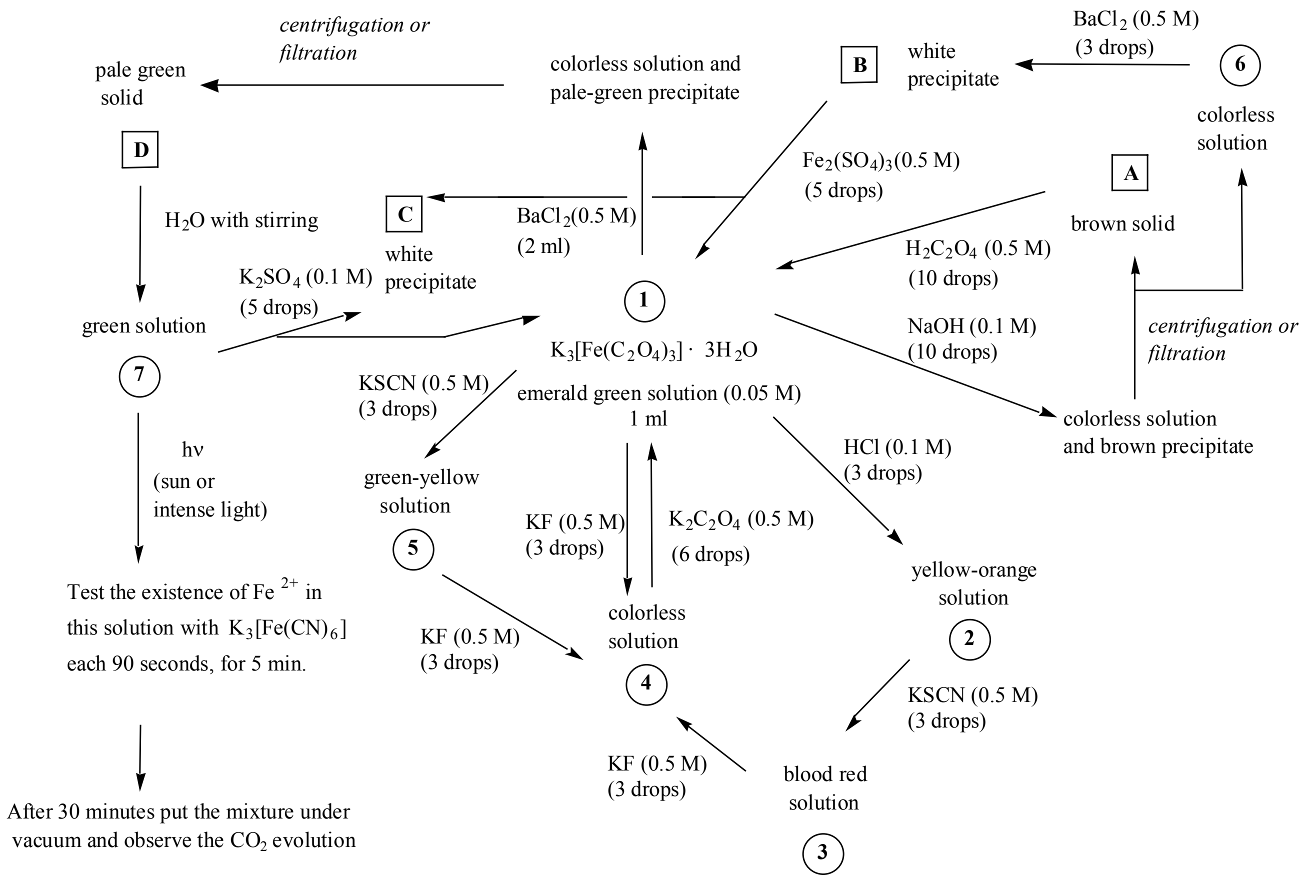
Supplementary Scheme 1



Supplementary Scheme 2



Supplementary Scheme 3



Supplementary Table 1

Reagent (concentration)	Laboratory material
KSCN (0.5 M)	Light (flashlight or 60-W bulb)
KF (3 M)	Test tube rack
$K_2C_2O_4$ (2 M)	Benchtop centrifuge and tubes (or funnel and filtration paper)
NaOH (0.1 M)	Vacuum (water aspirator pump)
HCl (6 M)	Graduated test tube (1-mL marks)
$H_2C_2O_4$ (1 M)	Pasteur pipettes
$Fe_2(SO_4)_3$ (1 M)	50-mL Beaker
$BaCl_2$ (0.2 M)	100-mL Kitasato flask
K_2SO_4 (0.1 M)	Test tube heating device
Distilled water	

CAS Registry numbers

KSCN	[*333-20-0*]
KF	[*7789-23-3*]
$K_2C_2O_4 \cdot H_2O$	[*6487-48-5*]
NaOH	[*1310-73-2*]
HCl	[*7647-01-0*]
$H_2C_2O_4 \cdot 2H_2O$	[*6153-56-6*]
$Fe_2(SO_4)_3 \cdot xH_2O$	[*15244-10-7*]
$BaCl_2 \cdot 2H_2O$	[*10326-27-9*]
K_2SO_4	[*7778-80-5*]